

IoT-Based Greenhouse System with AI Agent Consultation

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Abstract—Modern agriculture relies heavily on controlled environments like greenhouses to maximize crop yield. However, conventional monitoring systems often output raw telemetry that is difficult for non-technical operators to interpret, lacking long-term data retention and actionable intelligence. This paper presents an IoT-enabled smart greenhouse platform optimized for Cili cultivation, integrating edge computing, cloud data logging, and a Retrieval-Augmented Generation (RAG) Artificial Intelligence agent to improve diagnostic accessibility.

The proposed system utilizes an ESP32 microcontroller to continuously harvest environmental metrics, including ambient temperature, humidity, and soil moisture. Data is transmitted via the lightweight HiveMQ MQTT protocol to a Node-RED orchestration layer. The Node-RED middleware simultaneously logs the telemetry into a Firebase Realtime Database for permanent historical retention and queries an ElevenLabs Large Language Model (LLM) API. Experimental results demonstrate that the system successfully translates raw sensor variables into real-time, natural-language botanical consultations on a web dashboard, providing a highly stable and intelligent framework for sustainable precision agriculture.

Keywords—IoT; Smart Greenhouse; MQTT; Node-RED; AI Agent; LLM.

Moreover, data intelligence is lacking in conventional monitoring systems. Basic sensor networks show raw numerical readings, but they are unable to decipher the significance of those figures for plant health. Responses to crop stress are often delayed due to this communication gap between raw telemetry data and human comprehension. Large Language Models (LLMs), a type of artificial intelligence (AI), have recently drawn interest in smart agriculture as a way to close this gap by producing contextual, expert-level guidance from raw data inputs.

This paper presents an IoT-Based Smart Greenhouse System that integrates real-time sensor monitoring, cloud data logging, and an autonomous AI botanical agent. The proposed system utilizes an ESP32 microcontroller, MQTT communication, a Firebase Realtime Database, and the ElevenLabs API to support automated monitoring and generate natural-language diagnostic notices. The system is designed as a highly stable, data-driven solution for small-scale precision agriculture..

I. INTRODUCTION

In precision agriculture, maintaining a perfect crop environment is crucial to optimizing both output and health. The purpose of greenhouses is to maximize environmental parameters including soil moisture, temperature, and humidity. However, physical monitoring and visual observation play a major role in traditional farming setups, which results in uneven environmental management. Long-term analytical surveillance is impossible because farmers frequently operate blindly without real-time data logging or historical data preservation, even when they use simple automated setups.

Through the integration of microcontrollers, sensors, and wireless connectivity, recent developments in the Internet of Things (IoT) have made real-time monitoring and automation possible. Critical environmental parameters can be continuously monitored by incorporating IoT into greenhouses. When paired with dashboard platforms like Node-RED, lightweight communication protocols like MQTT provide effective data transfer and real-time viewing with low network overhead.

II. RELATED WORK

The application of Internet of Things (IoT) technology in smart farming has gained increasing attention due to its ability to improve monitoring accuracy and resource efficiency. Several studies have explored sensor-based greenhouse systems that focus on real-time monitoring of environmental parameters. Simo et al. (2022) developed a baseline sensor-to-dashboard tracking architecture utilizing edge microcontrollers for temperature and humidity regulation. Their work emphasizes that automated monitoring improves operational reliability in controlled agricultural spaces..

Other studies have investigated the integration of actuators for physical automation. Mishra (2023) and Manoharan et al. (2024) proposed IoT-enabled greenhouses that utilize sensors to collect data and trigger actuators such as fans and pumps to optimize plant growth. While these systems demonstrate that IoT can significantly reduce manual intervention, they rely on rigid, rule-based threshold logic. Furthermore, these systems often operate without permanent cloud data logging, limiting their predictive capabilities and long-term data analysis..

Artificial intelligence and Retrieval-Augmented Generation (RAG) frameworks have also been introduced to support automated observation and complex decision-making. Karim et al. (2025) demonstrated that AI agents combined with knowledge-retrieval mechanisms can process complex datasets and generate informed responses grounded in verified information. These approaches enable highly adaptive decision-making when processed externally via APIs.

Existing smart greenhouse systems often handle IoT-based monitoring, hardware automation, or AI-assisted analysis as distinct components, according to the studied literature. There hasn't been much study done on combining generative AI diagnostics, persistent NoSQL cloud logging (Firebase), and real-time MQTT telemetry on one platform. In order to close this gap, this study presents a system that combines Node-RED visualization with an LLM agent to provide comprehensible agronomy advice while giving software stability and data translation first priority.

III. METHODOLOGY

The proposed IoT-Based Smart Greenhouse System is designed to provide automated monitoring and AI-driven diagnostics through the integration of edge sensor data acquisition, wireless communication, and cloud orchestration. The overall system architecture consists of three main components: the edge sensing unit, the cloud middleware, and the AI intelligence layer.

An ESP32 microcontroller is used as the central edge controller due to its built-in Wi-Fi capability and suitability for IoT applications. Environmental data is collected using a DHT22 sensor (temperature and humidity) and an analog soil moisture sensor. During the prototyping phase, it was determined that the inductive load of low-cost physical water pumps caused severe electromagnetic interference (EMI), leading to sensor data corruption. Consequently, physical water actuators were excluded to prioritize 100% data integrity, with the system relying on the dashboard AI for hydration alerts. An electrically isolated L9110 DC exhaust fan is retained for automated thermal regulation.

The collected data is transmitted using the MQTT protocol (HiveMQ) to a Node-RED orchestration layer. Node-RED serves as the central middleware, providing a real-time visualization dashboard. Node-RED simultaneously formats the incoming telemetry into JSON payloads and executes HTTP POST requests to a Firebase Realtime Database for permanent historical data retention.

In addition to sensor-based monitoring and logging, Node-RED acts as a bridge to the intelligence layer. When significant environmental fluctuations occur, the JSON payload is transmitted to the ElevenLabs API. An expert botanical AI agent processes the numerical variables to generate natural-language consultation text, which is routed back and displayed directly on the Node-RED UI.

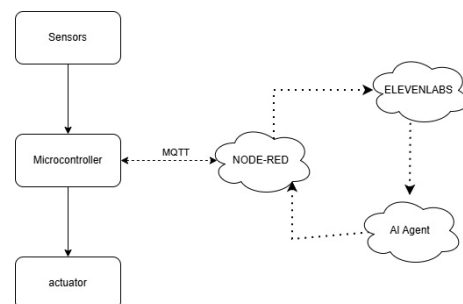


Fig. 1. Block diagram of the IoT-based solar-powered smart aquaponics system.

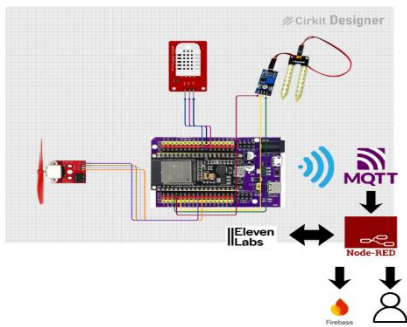


Fig. 2. System architecture and physical layout of the IoT-based smart aquaponics system.

IV. RESULT AND DISCUSSION

The created IoT-based smart greenhouse system was assessed to confirm the AI consultation agent's operational effectiveness, system stability, and monitoring accuracy. Sensor reliability, data logging, thermal automation, and natural-language text synthesis were all tested in simulated environments.

Real-time telemetry from the DHT22 and soil moisture sensors was successfully collected by the ESP32 edge architecture. The system showed remarkable electrical stability with no data corruption or voltage drops when the EMI-prone water pump was removed. The MQTT protocol was successfully used to transmit real-time sensor data, which were then shown on the Node-RED dashboard. Through the use of radial gauges and historical line charts, the dashboard offered constant, clear representation of environmental information..

The cloud database synchronization was verified in order to assess the data logging capabilities. The MQTT telemetry was successfully converted into structured JSON objects by the Node-RED runtime and uploaded to the Firebase Realtime Database. The findings establish a trustworthy historical dataset for upcoming analytics by confirming that each sensor update was permanently recorded without packet loss.

The local thermal control loop, which was calibrated for Cili cultivation, was evaluated to test the automation of the system. The ESP32 successfully activated the relay module to power the L9110 exhaust fan when fake heat was provided to the DHT22 sensor, raising the temperature above the crucial 34°C threshold. Concurrently, the "FAN STATUS" indicator on the Node-RED dashboard was changed to "ON" by the MQTT broker, indicating that the edge-to-cloud synchronization was successful.

Finally, the AI botanical agent was evaluated for diagnostic accuracy. When the soil moisture sensor was manually removed from the soil to simulate a critical drop below 20%, Node-RED successfully queried the ElevenLabs API. The AI agent accurately processed the variables and generated a natural-language warning detailing the risks of low biomass accumulation, advising immediate hydration. The text was successfully rendered on the dashboard alongside a high-priority UI toast notification..

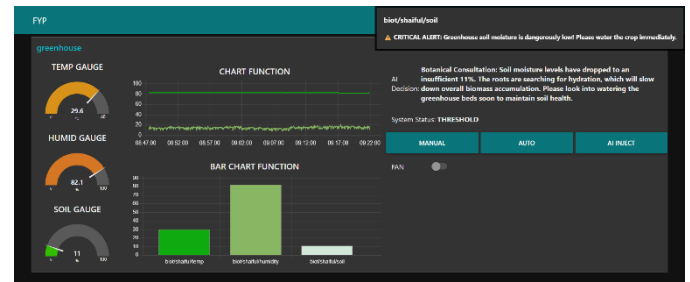


Fig. 3. Real-time monitoring dashboard implemented using Node-RED

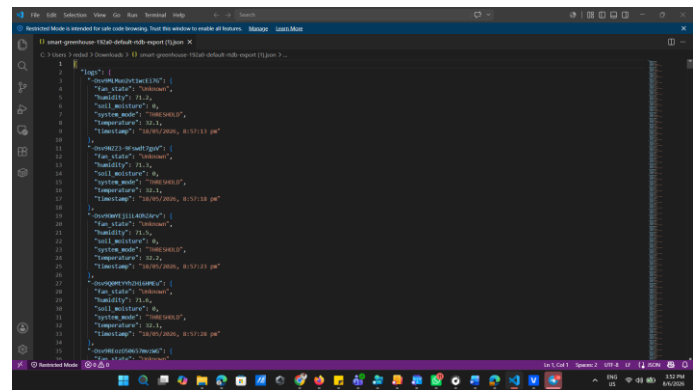


Fig. 4. Firebase Realtime Database JSON logs

V. CONCLUSION

This paper presented an IoT-Based Smart Greenhouse System designed to support automated monitoring, robust data retention, and diagnostic accessibility for precision agriculture. The proposed system integrates real-time edge sensing, MQTT-based data communication, Node-RED visualization, and a Firebase Realtime Database to enable continuous cloud logging.

Experimental evaluation demonstrated that the system is capable of reliably monitoring key environmental parameters, including temperature, humidity, and soil moisture. By strategically excluding unstable physical water actuators, the system achieved zero data corruption. The use of Node-RED provided effective real-time visualization and data orchestration, while MQTT communication ensured stable data transmission with minimal latency. Furthermore, the integration of a Retrieval-Augmented Generation (RAG) AI agent successfully bridged the gap between raw hardware metrics and human understanding, automatically translating telemetry into expert-level agricultural advice directly on the dashboard.

Overall, the proposed system demonstrates a practical and scalable approach for integrating IoT, NoSQL cloud databases, and Large Language Models in smart agriculture. It provides a highly stable foundation for future enhancements, such as optically-isolated autonomous irrigation, enterprise-level API orchestration via n8n, and the application of predictive machine learning for time-series crop forecasting.

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